THE RANGE-WEAK-COMPACTNESS QUANTIFICATION OF GROTHENDIECK IMPLIES BENDOVA'S QUANTIFICATION

AGENT LANE 10

Source and Question

The source paper is Hana Krulisova, *C*-algebras have a quantitative version of Pelczynski's property (V)*, arXiv:1605.04900. In Section 5, after recalling Bendova's quantitative Grothendieck property, the paper introduces a second possible quantification:

$$\operatorname{wck}_{X'}\big(\{x'_n : n \in \mathbb{N}\}\big) \le c\, \delta_{w^*}(x'_n)$$

for every bounded sequence (x'_n) in X'. The paper then asks whether this "latter quantitative Grothendieck property implies the former one (with a larger constant)." The former one is Bendova's inequality

$$\delta(x'_n) \le C\, \delta_{w^*}(x'_n)$$

for every bounded sequence in X'.

Proof Intuition

The only possible obstruction is that weak compactness of the range is a property of the set $\{x'_n\}$, while $\delta(x'_n)$ depends on the order of the sequence. The gap is bridged by looking at two weak-star cluster points $u, v \in X'''$ obtained from two far-out subsequences of (x'_n). Their difference has two parts. The part detected on the canonical copy of X is controlled exactly by $\delta_{w^*}(x'_n)$. The remaining part annihilates X, and an elementary quotient estimate bounds it by the distances of u and v from the canonical copy of X', hence by the weak non-compactness measure $\operatorname{wck}_{X'}(\{x'_n\})$.

Notation

For a bounded sequence (x'_n) in X', set

$$\delta(x'_n) = \sup_{x'' \in B_{X''}} \lim_{m\to\infty} \sup_{k,\ell \ge m} |x'_k(x'') - x'_\ell(x'')|$$

and

$$\delta_{w^*}(x'_n) = \sup_{x \in B_X} \lim_{m\to\infty} \sup_{k,\ell \ge m} |x'_k(x) - x'_\ell(x)|.$$

These are the quantities used in the source paper. We regard X' canonically as a subspace of X''' and write $J : X' \to X'''$ for this embedding. Let $\rho : X''' \to X'$ be the restriction map to the canonical copy of X in X'':

$$\rho(u)(x) = u(J_X x), \qquad u \in X''',\ x \in X.$$

Lemma 1. *If $h \in X'''$ satisfies $\rho(h) = 0$, then*

$$\|h\| \le 2\, \operatorname{dist}(h, JX').$$

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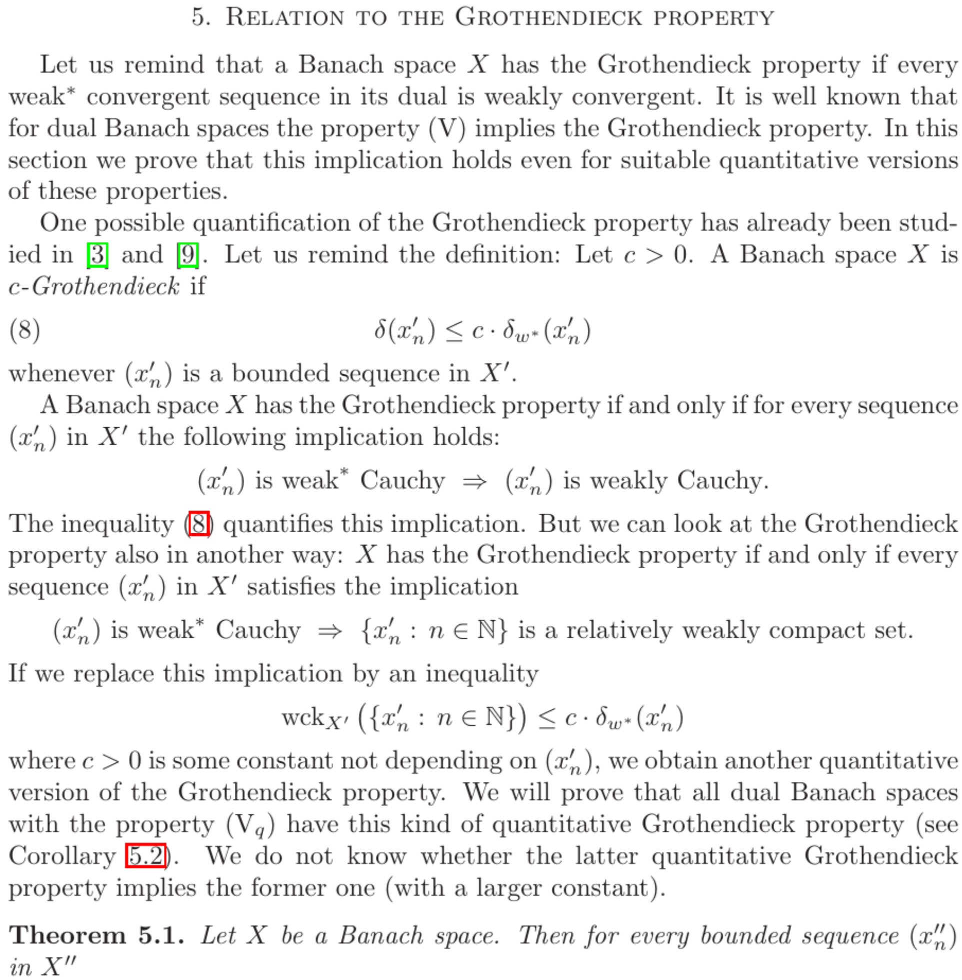

5. Relation to the Grothendieck property

Let us remind that a Banach space X has the Grothendieck property if every weak* convergent sequence in its dual is weakly convergent. It is well known that for dual Banach spaces the property (V) implies the Grothendieck property. In this section we prove that this implication holds even for suitable quantitative versions of these properties.

One possible quantification of the Grothendieck property has already been studied in [3] and [9]. Let us remind the definition: Let $c > 0$. A Banach space X is *c-Grothendieck* if

$$\delta(x'_n) \leq c \cdot \delta_{w^*}(x'_n) \tag{8}$$

whenever (x'_n) is a bounded sequence in X'.

A Banach space X has the Grothendieck property if and only if for every sequence (x'_n) in X' the following implication holds:

$$(x'_n) \text{ is weak}^* \text{ Cauchy } \Rightarrow (x'_n) \text{ is weakly Cauchy.}$$

The inequality (8) quantifies this implication. But we can look at the Grothendieck property also in another way: X has the Grothendieck property if and only if every sequence (x'_n) in X' satisfies the implication

$$(x'_n) \text{ is weak}^* \text{ Cauchy } \Rightarrow \{x'_n : n \in \mathbb{N}\} \text{ is a relatively weakly compact set.}$$

If we replace this implication by an inequality

$$\operatorname{wck}_{X'}\left(\{x'_n : n \in \mathbb{N}\}\right) \leq c \cdot \delta_{w^*}(x'_n)$$

where $c > 0$ is some constant not depending on (x'_n), we obtain another quantitative version of the Grothendieck property. We will prove that all dual Banach spaces with the property (V_q) have this kind of quantitative Grothendieck property (see Corollary 5.2). We do not know whether the latter quantitative Grothendieck property implies the former one (with a larger constant).

Theorem 5.1. *Let X be a Banach space. Then for every bounded sequence (x''_n) in X''*

Figure 1. The source passage on page 9 asking whether the range-weak-compactness quantification implies Bendova's quantitative Grothendieck property.

Proof. Let $y' \in X'$. Since $\rho(h) = 0$, the restriction of $h - Jy'$ to X is $-y'$. Hence

$$\|y'\| \leq \|h - Jy'\|.$$

Therefore

$$\|h\| \leq \|h - Jy'\| + \|Jy'\| = \|h - Jy'\| + \|y'\| \leq 2\|h - Jy'\|.$$

Taking the infimum over $y' \in X'$ gives the result. $\square$

Theorem 1. *Let X be a Banach space and let (x'_n) be a bounded sequence in X'. Put $A = \{x'_n : n \in \mathbb{N}\}$. Then*

$$\delta(x'_n) \leq \delta_{w^*}(x'_n) + 4 \operatorname{wck}_{X'}(A).$$

Proof. Let

$$r = \operatorname{wck}_{X'}(A), \qquad d = \delta_{w^*}(x'_n).$$

Fix $x'' \in B_{X''}$. We show that the tail oscillation of the scalar sequence $(x'_n(x''))$ is at most $d+4r$.

Let L denote this tail oscillation. If $L = 0$ there is nothing to prove. For $\varepsilon > 0$, choose subsequences (x'_{p_j}) and (x'_{q_j}) with $p_j, q_j \to \infty$ such that

$$|x'_{p_j}(x'') - x'_{q_j}(x'')| > L - \varepsilon$$

along a further scalar-convergent choice. By weak-star compactness in X''', pass to weak-star cluster points u of (x'_{p_j}) and v of (x'_{q_j}) so that

$$|u(x'') - v(x'')| \geq L - \varepsilon.$$

Both u and v are weak-star cluster points of sequences in A, so by the definition of wck,

$$\operatorname{dist}(u, JX') \leq r, \qquad \operatorname{dist}(v, JX') \leq r.$$

For each $x \in B_X$, the two numbers $u(J_X x)$ and $v(J_X x)$ are cluster values of the scalar sequence $(x'_n(x))$ along tails. Hence

$$|\rho(u - v)(x)| \leq d.$$

Thus $\|\rho(u - v)\| \leq d$.

Set

$$h = (u - v) - J\rho(u - v).$$

Then $\rho(h) = 0$. Moreover, since $J\rho(u - v) \in JX'$,

$$\operatorname{dist}(h, JX') = \operatorname{dist}(u - v, JX') \leq \operatorname{dist}(u, JX') + \operatorname{dist}(v, JX') \leq 2r.$$

Lemma 1 gives $\|h\| \leq 4r$. Therefore

$$\begin{aligned} L - \varepsilon &\leq |u(x'') - v(x'')| \\ &\leq |J\rho(u - v)(x'')| + \|h\| \\ &= |x''(\rho(u - v))| + \|h\| \\ &\leq d + 4r. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ and then taking the supremum over $x'' \in B_{X''}$ proves the theorem. □

Corollary 1. *Suppose that X satisfies the source paper's range-weak-compactness quantitative Grothendieck property with constant c, i.e.*

$$\operatorname{wck}_{X'}\big(\{x'_n : n \in \mathbb{N}\}\big) \leq c\, \delta_{w^*}(x'_n)$$

for every bounded sequence (x'_n) in X'. Then X is $(1 + 4c)$-Grothendieck in Bendova's sense:

$$\delta(x'_n) \leq (1 + 4c)\, \delta_{w^*}(x'_n)$$

for every bounded sequence (x'_n) in X'.

Proof. Apply Theorem 1 and substitute the assumed bound for $\operatorname{wck}_{X'}(\{x'_n : n \in \mathbb{N}\})$. □

Consequence for the Source Paper

This gives an affirmative answer to the question asked on page 9 of arXiv:1605.04900: the second quantitative Grothendieck property does imply Bendova's quantitative Grothendieck property, with an explicit larger constant. In particular, combining this packet with Corollary 5.2 of the source paper, any dual Banach space whose dual has the quantitative property (V_q) is quantitatively Grothendieck in Bendova's sense with the corresponding enlarged constant. For a von Neumann algebra, the source paper gives

$$\operatorname{wck}_{A'}(\{x'_n : n \in \mathbb{N}\}) \leq \frac{\pi}{2}\, \delta_{w^*}(x'_n),$$

so the present result yields the Bendova inequality with constant $1 + 2\pi$.

Limitations

This packet does not settle the separate question, asked earlier in the same paper, whether all C^*-algebras have the property $(V_q)_{\omega^*}$, nor the broader question whether every Banach space with (V_q) has $(V_q)_{\omega^*}$. The result here is only about the two quantitative Grothendieck formulations in Section 5.

Novelty Check

Before promotion, the run checked the registry, solution, attempt, and proof-gap lightweight indexes for arXiv:1605.04900 and the phrases "quantitative Grothendieck", "delta_w*", "wck", "Pelczynski property (V)", and related variants. No duplicate packet for this question was found. Web searches for exact phrases including "delta-w-star wck Grothendieck Krulisova", "wck delta-w-star delta-w Grothendieck", and "weak-star non-Cauchyness wck Grothendieck property" did not reveal a later paper answering the bridge question. The local parsed corpus contained the source paper, Bendova's definition as cited in later surveys, and related Grothendieck-property papers, but no instance of the inequality in Theorem 1.

References

[1] H. Krulisova, *C*-algebras have a quantitative version of Pelczynski's property (V)*, arXiv:1605.04900.
[2] H. Bendova, *Quantitative Grothendieck property*, J. Math. Anal. Appl. 412 (2014), 1097–1104.